

Geotechnical Monitoring

The Observational Method in Design

In the 1940's Karl Terzaghi adapted the phenomenological approach to develop a systematic means to solve geotechnical problems. This has become known as the "observational method", the conceptualization behind which Terzaghi wrote (paraphrased here):

"In the engineering of large geotechnical works, a vast amount of effort goes towards securing only *roughly approximate* values for the physical constants that appear in the equations. In these equations many additional variables are not considered or remain unknown. Therefore, the results of computations are no more than *working hypotheses*, subject to confirmation or modification during construction."

The Observational Method in Design

"In the past, only two methods have been used for coping with inevitable uncertainties: either to adopt an *excessive factor of safety*, or else to make assumptions in accordance with *general experience*. The first of these is wasteful; the second is dangerous as most failures occur due to unanticipated or unknown geological factors/processes."

"A third method, *the observational method*, provides a „learn as you go“ alternative. The procedure for this is to base the design on whatever information can be secured, make note of all possible differences between reality and the assumptions, then compute, on the basis of the original assumptions, various quantities that can be measured in the field. Based on the results of these measurements, gradually close the gaps in knowledge and, if necessary, *modify the design during construction*."

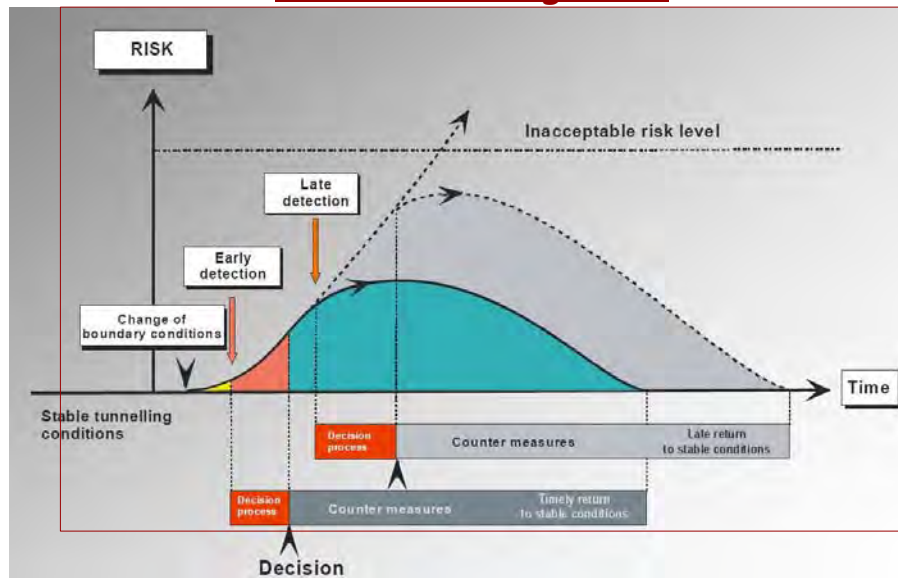
Terzaghi & Peck (1948)

Geotechnical Monitoring

Monitoring provides feedback on the design process. If the design is stable, monitoring can help to assess how conservative the design was and whether there is scope for optimization. If the designed rock engineering structure of interest is not stable, then monitoring can provide early warning and an opportunity to control the consequences of the failure. Therefore *monitoring has the dual purpose of optimizing the rock engineering design and assists in controlling the risk of failure*. Once the rock mass response is understood, monitoring can also improve the capabilities to forecast the most likely outcome of a design based on early readings.

Monitoring is much more powerful if it is used systematically as part of the design process rather than sporadically when there is a perceived increased risk at a certain location. Simple means of monitoring like visual observation can be powerful if the data is well documented, compiled and analysed regularly. Instrumentation data however can provide measurable outcomes and assist in understanding rock mass behaviour and how it responds to mining.

Geo-Risk Management



Instrumentation & Monitoring

The use of geotechnical instrumentation is not merely the selection of instruments but a comprehensive step-by-step engineering process beginning with a definition of the **objective** and ending with **implementation** of the data.

Engineering objectives typically encountered in soil and rock engineering projects have led to the design and **commercial marketing** of numerous instrument types, measuring for example:

- decreasing reliability ↓
- temperature
 - deformation
 - groundwater/pore pressures
 - total stress in backfill and stress change in rock



Instrumentation. Monitoring & Tunnel Design

The required versatility in how instruments can be deployed (on surface, from boreholes, etc.) and what they are meant to measure (rock properties, ground movements, water pressures, etc.) has led to the development of a wide variety of devices.

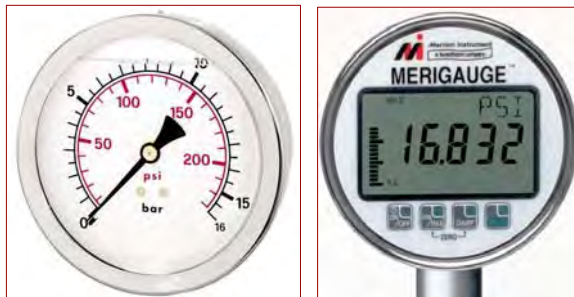
When choosing instruments for a particular project, the engineer must consider and balance the job-related requirements of:

Range - the maximum distance over which the measurement can be performed, with greater range usually being obtained at the expense of resolution.



Instrumentation. Monitoring & Tunnel Design

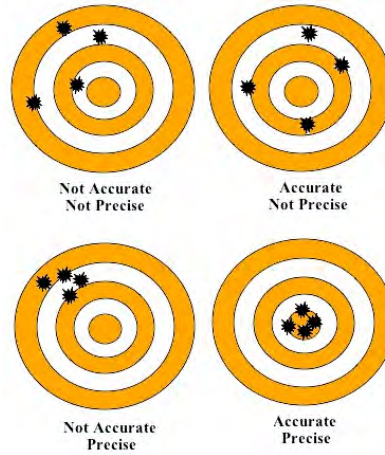
Resolution - the smallest numerical change an instrument can measure.



Instrumentation, Monitoring & Tunnel Design

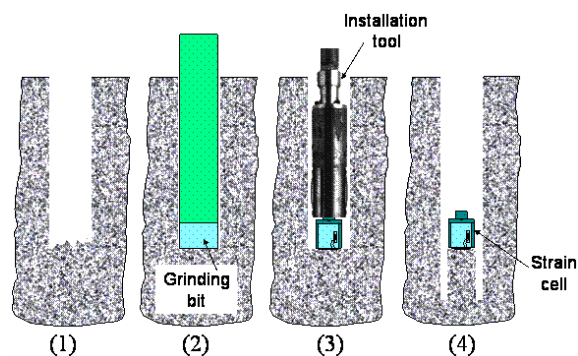
Accuracy - the degree of correctness with respect to the true value, usually expressed as a $\pm$ number or percentage.

Precision - the repeatability of similar measurements with respect to a mean, usually reflected in the number of significant figures quoted for a value.



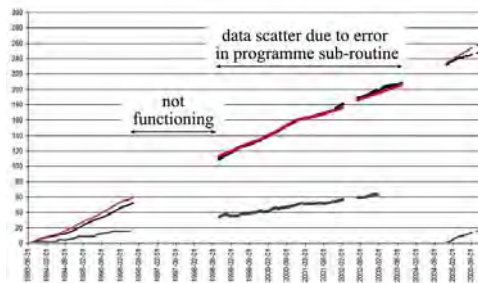
Instrumentation, Monitoring & Tunnel Design

Conformance - whether the presence of the instrument affects the value being measured.



Instrumentation, Monitoring & Tunnel Design

Robustness - the ability of an instrument to function properly under harsh conditions to ensure data accuracy and continuity are maintained.



Reliability - synonymous with confidence in the data; poor quality or inaccurate data can be misleading and is worse than no data.

Field Instrumentation

Tunnel projects often present the ultimate measurement challenge, in part because of their initial **lack of definition** and the sheer **scale of the problem**. The measurement problem usually requires information ranging from a **coarse scale** down to a **fine scale** and involving a number of instrumentation techniques.

The ultimate goal is to select the most **sensitive measurement parameters** with respect to the **project objectives**, e.g. the ones that will change significantly at the onset of failure. However, because of **physical limitations** and **economic constraints**, all parameters cannot be measured with equal ease and success.

Warning Level	Criterion	Action
1	Movement greater than 10 mm at any one survey station	Report to mine management
2	Movement greater than 15 mm at two adjacent stations; or velocity exceeding 15 mm per month at any one station	Verbal report and site meeting followed by written report and recommendations
3	Movement greater than 15 mm plus acceleration at any one station	Immediate site inspection by consulting engineer, site meeting and probable remedial measures (according to contingency plans)

Franklin (1977)

Planning and Design

Predictive monitoring systems are best implemented after a period of **investigative monitoring**. Monitoring information must be assessed in the context of the **physical setting** and the conclusions of the **investigation phase**.

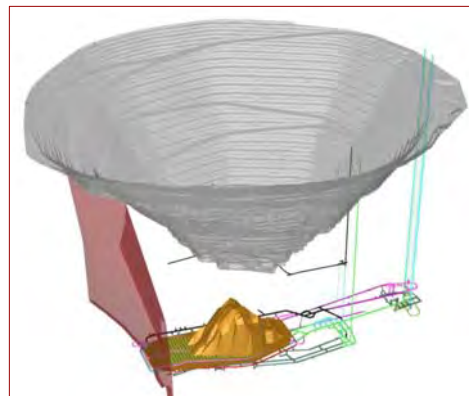
Planning of a monitoring program should be **logical and comprehensive** as the measurement problem may require a number of **different** instrument types collecting information across a range of **varying scales**. Furthermore, because of physical limitations and economic constraints, all parameters cannot be measured with equal ease and success.

Planning and Design

Adequate planning is required before proceeding, and these plans should be logical and comprehensive - from defining the objectives to planning how the measurement data will be implemented.

1. Define the project conditions.

... these might include project type and layout, geology and structure, engineering properties of materials, groundwater conditions, status of nearby structures or other facilities, environmental conditions, and planned construction method.

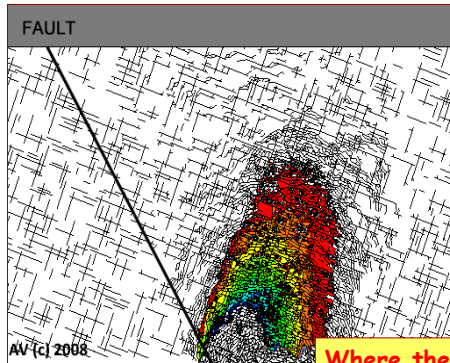


Woo et al. (2012)

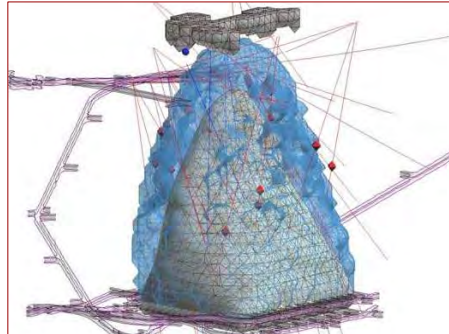
Planning and Design

2. Predict mechanisms that control behaviour.

... may involve one or more working hypotheses based on a comprehensive knowledge of the project conditions.



Vyazmensky et al. (2010)



Where the rock mass involves brittle fracturing, microseismic monitoring should form an integral part of the monitoring strategy.

Planning and Design

3. Define the questions that need to be answered.

... every instrument on a project should be selected and placed to assist in answering a specific question: if there is no question, there should be no instrumentation.



(i)Investigative Monitoring: To provide an understanding of the rock mass behaviour and thus enable appropriate actions to be implemented.

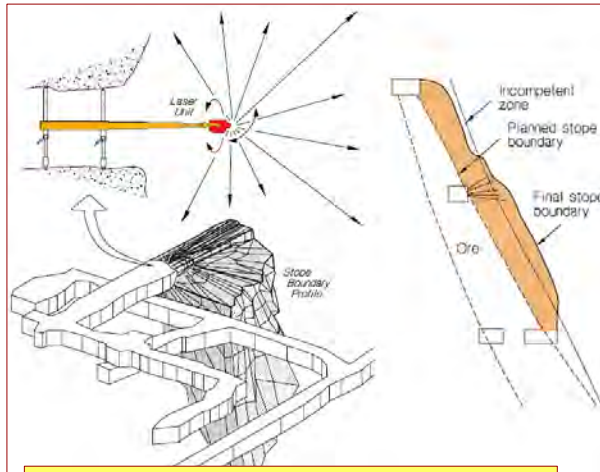
(ii)Predictive Monitoring: To provide a warning of a change in behaviour and thus enable the possibility of limiting damage or intervening to prevent failure.

Planning and Design

Hutchinson & Diederichs (1996)

4. Define purpose of the instrumentation.

... Peck (1984) stated that, "The legitimate uses of instrumentation are so many, and the questions that instruments and observation can answer so vital, that we should not risk discrediting their value by using them improperly or unnecessarily."



Use of laser distance meter at Hemlo gold mine to monitor hangingwall dilution.

Planning and Design

5. Select the parameters to be monitored.

... Parameters include pore water pressure, microseismicity, stress change, deformation, and load and strain in structural members.

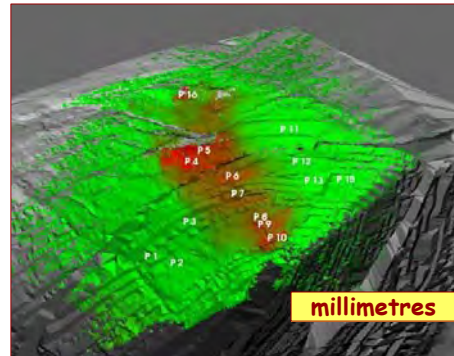


... high precision joint water pressure measurements from a borehole in the Bedretto tunnel adit, Switzerland.

Planning and Design

6. Predict magnitudes of change.

... Predictions are necessary so that the required instrument ranges and required instrument sensitivities or accuracies can be selected.



Severin et al. (2011)

Planning and Design

7. Devise remedial action.

... Inherent in the use of instrumentation for construction purposes is the absolute necessity for deciding in advance, a positive means for solving any problem that may be disclosed by the results of the observation.

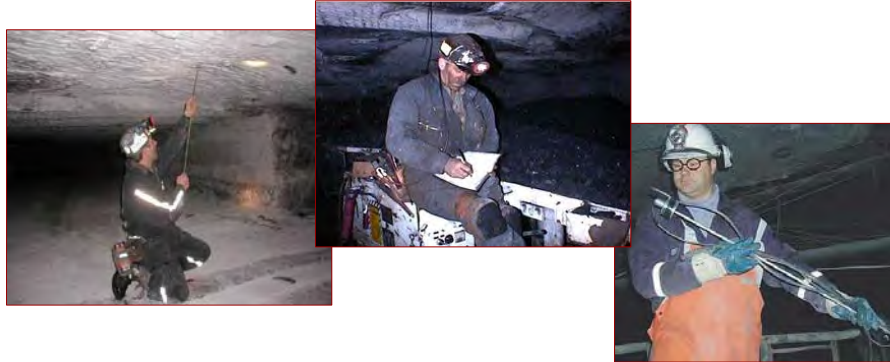


- critical alarm situation; emergency is announced and mine/tunnel superintendent is notified to evacuate.
- geotech alarm situation; movements indicate developing situation that geotech department should provide guidance on.
- system failure; mine/tunnel superintendent notified that monitoring system is unavailable and geotech department notified to assess problem.
- all systems go and movements below alarm thresholds.

Planning and Design

8. Assign tasks for design, construction and operation phases.

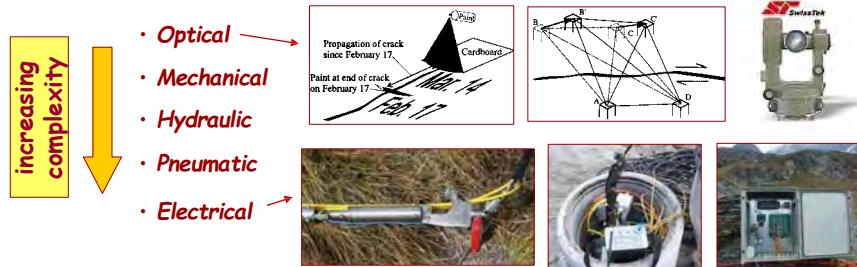
... Instrumentation specialists may be mine/tunnel staff or outside consultant/ contractor with special expertise. When assigning tasks for monitoring, the party with the greatest vested interest in the data should be given direct line responsibility for producing it accurately.



Planning and Design

9. Select Instruments.

... When selecting instruments, the overriding requirement is **reliability!**
Inherent in reliability is simplicity.



Instruments should be: ... reliable, rugged and capable of functioning for long periods of time without repair or replacement;

... capable of responding rapidly and precisely to changes so that a true picture of events can be maintained at all times.

Planning and Design - Sensor Types

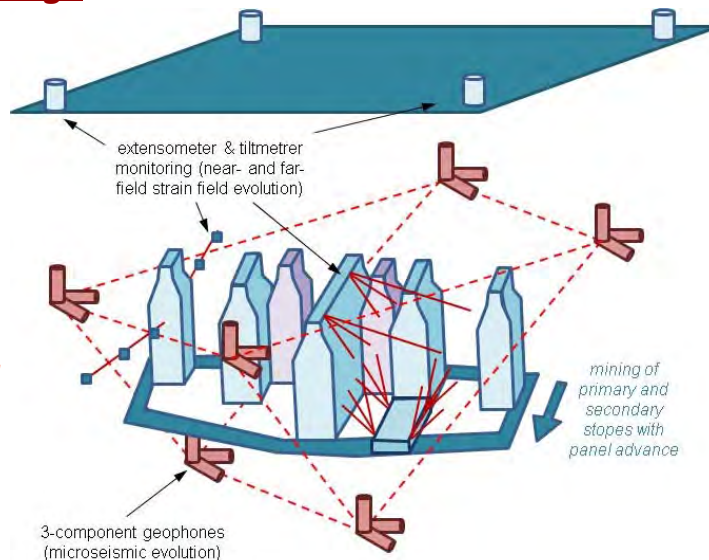
Eberhardt & Stead (2011)

Sensor Type	Operation Principles
LVDT	Linear Variable Differential Transformers (LVDT) consist of a movable magnetic core passing through one primary and two secondary coils. An excitation voltage is applied and a voltage is induced in each secondary coil. When the core moves off center, the output voltage increases linearly in magnitude. Commonly used in instruments to measure displacements.
Vibrating wire	Involves a high tensile steel wire fixed at both ends and tensioned so that it is free to vibrate at its natural frequency. The wire is magnetically plucked by an electrical coil and its frequency measured. When one end moves relative to the other the tension in the wire, and therefore measured frequency, changes. Commonly used in pressure cells, piezometers and deformation gages.
Accelerometers	Consist of a damped mass suspended in a magnetic field; under the influence of external accelerations (or motion) the mass deflects from its neutral position and the deflection measured. Commonly used in tiltmeters and inclinometers.
Fiber optics	Light is emitted into and confined to a glass fiber core and propagates along the length of the fiber. Any disturbance of the fiber alters the guided light which can then be related to the magnitude of the disturbing influence. Finding increased use in piezometers and deformation monitoring instruments.
MEMS	Micro-Electrical Mechanical Systems (MEMS) are small integrated devices that combine electrical and mechanical components on a sub-millimeter scale. This allows for transducers that are much smaller, more functional, lighter, more reliable, and produced for a fraction of the cost of other transducers.

Planning and Design

10. Select instrument locations.

... The selection of instrument locations should reflect predicted behaviour and should be compatible with the method of analysis that will later be used when interpreting the data.



Planning and Design

11. Account for factors that may influence measured data.

... Details of each instrument installation should be recorded, because local or unusual conditions often influence measured variables.

12. Establish procedures for ensuring reading correctness.

... When reading an instrument, one should be able to answer: Is the instrument functioning correctly? The answer can be provided by visual observations, duplication of instruments, data consistency or through the use of instruments that internally check their own correct functioning.

Instrument Environments

1. Large deformations—often shearing deformations
2. High pressures—both solids and fluids
3. Corrosive—chemical (groundwater, grouts, concrete additives, bacteria) and electrolytic (electrolysis of dissimilar materials, stray electrical currents)
4. Temperature extremes—subfreezing to 100°F+ in the sun (temperature can be higher in certain instances, such as nuclear waste storage)
5. Shock—blasting, construction activities, rough handling during transportation to and from site
6. Vandalism, destruction by construction equipment, fly rock
7. Dust, dirt, mud, rain, chemical precipitates
8. High humidity, flowing or standing water
9. Erratic power supplies (electrical instruments)
10. Loss of accessibility to instruments when covered by rock, soil, shotcrete, and other supports

Planning and Design

13. List the specific purpose of each instrument.

... It is useful to question whether all planned instruments are justified. If no viable purpose can be found for a planned instrument, it should be deleted.

14. Prepare budget.

... A budget should be prepared for all instrumentation-related tasks to ensure sufficient funds are available. A frequent error in budgeting is to underestimate the data collection and processing costs and time commitments.

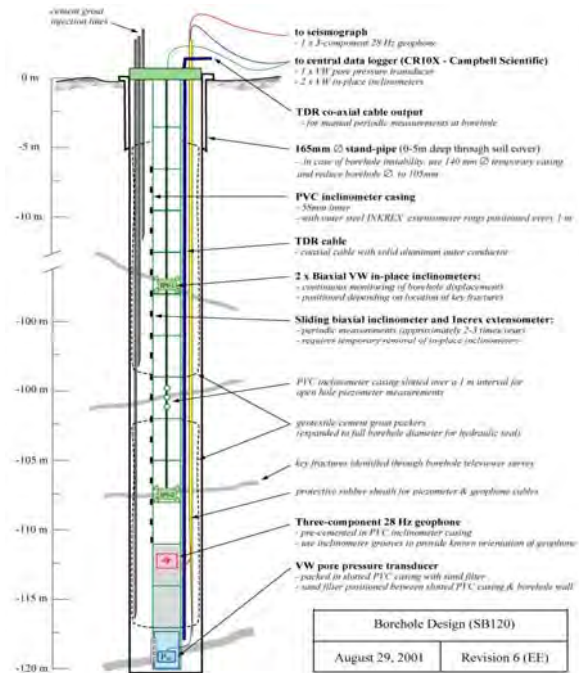
15. Prepare instrument procurement specifications.

... There are several competing instrumentation manufacturers, each offering products designed for similar geotechnical purposes. Be aware of an instrument's capabilities (and limitations), as well as those of competing products. Instruments should be purchased from established manufacturers.

Planning and Design

16. Plan installation, regular calibration & maintenance.

... Installation procedures should be planned well in advance of scheduled installation dates. The step-by-step procedures should include a detailed list of required materials and tools, and installation record sheets for documenting factors that may influence measured data.

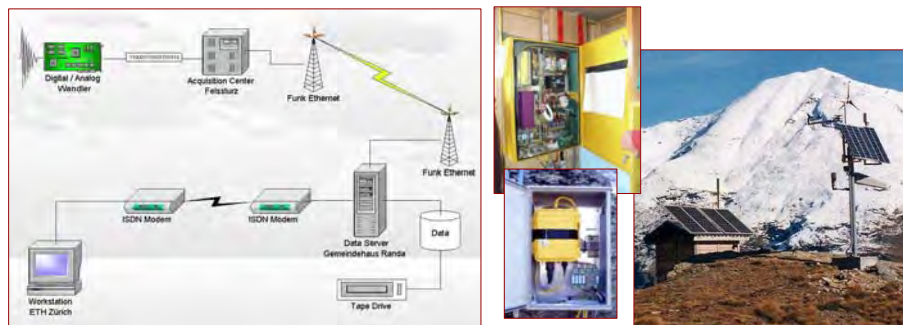


Eberhardt (2008)

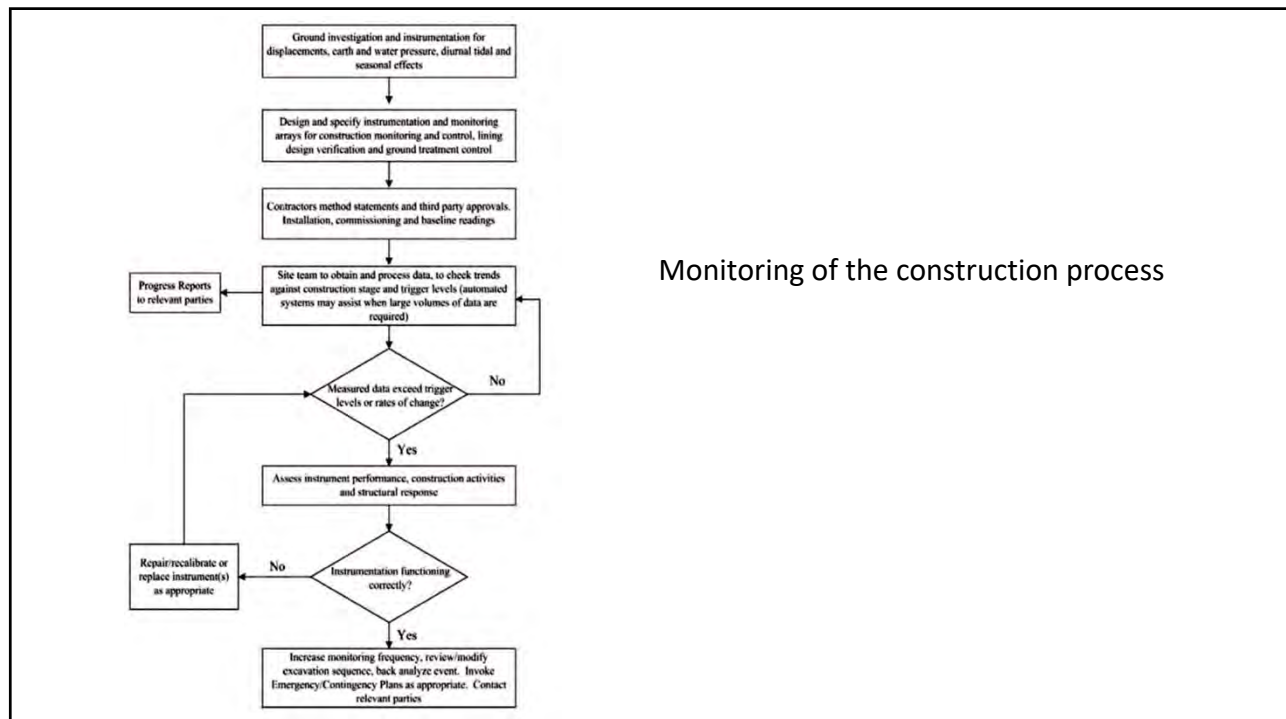
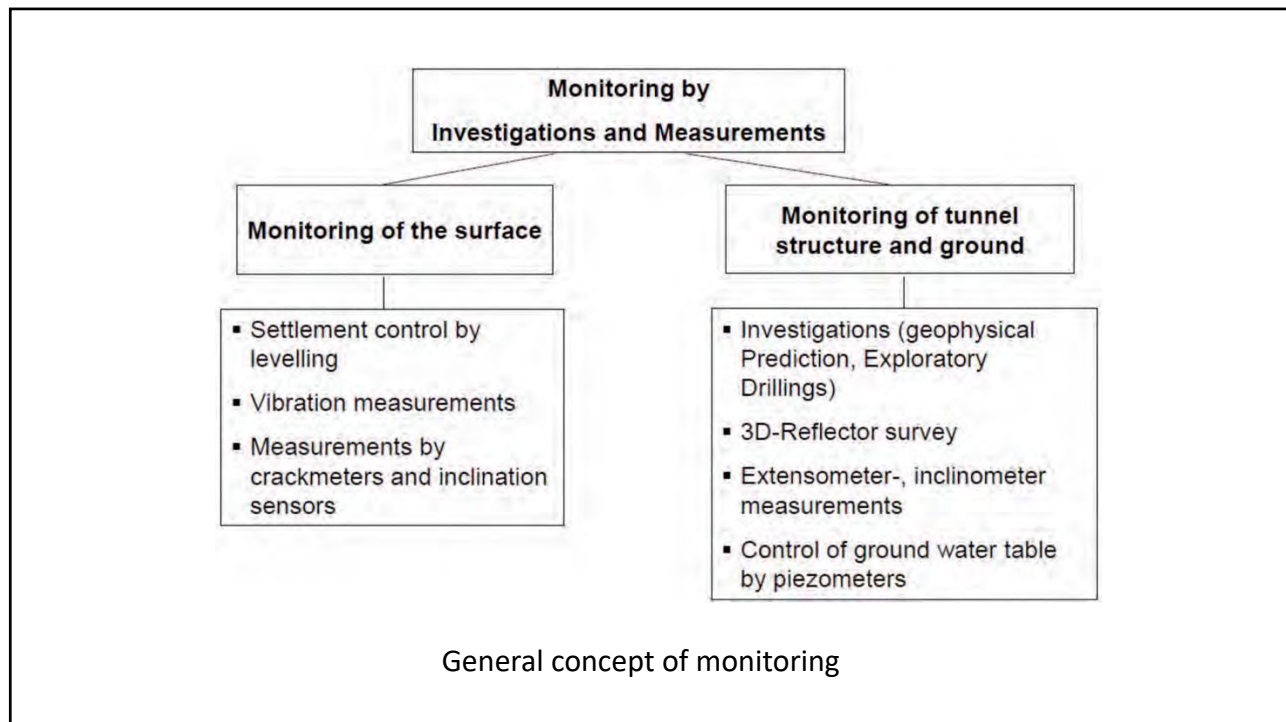
Planning and Design

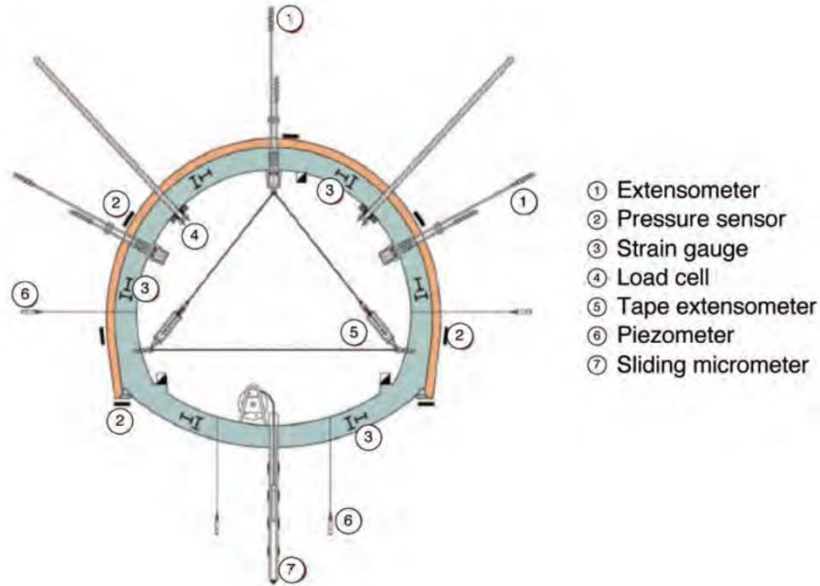
17. Plan data collection, processing, presentation, interpretation, reporting and implementation.

... The effort required for these tasks should not be underestimated. Many consulting firms have files filled with large quantities of partially processed and undigested data because sufficient time/funds were not available for these tasks.



Willenberg et al. (2008)





Typical layout of instrumentation

	Conventional Method (low overburden in urban areas)	Closed-face TBM (urban areas)	Operational tunnel in creeping ground
0. VISUAL INSPECTION (face & sidewall)	●	●	●
1. GEOMETRICAL PARAMETERS			
Face extrusion (horizontal displacement)	●		
Surface settlement	●	●	
Surface rotation	○	○	
Extrusion of the ground ahead of tunnel face (extrusion-meter)	■		
Displacement in borehole (extensometer, inclinometer)	○	○	×
Convergence at sidewall	●		●
Crack monitoring	○		●
Deformation of permanent lining	×	×	●
2. MECHANICAL PARAMETERS			
Force (arch base, anchoring rod, rock bolt, etc.)	●		×
Stress in ground			○
Stress in support /lining	○	×	○
3. HYDRAULIC PARAMETERS			
Pumped out water rate	○		○
Surface rainfall	×		×
Piezometric levels in ground	●	●	●
Temperature of leakage			×
4. OTHER PARAMETERS			
Tunnel air temperature		○	×
Tunnel air pressure		×	
Tunnel hygrometry	×	×	○
Date and time	●	●	●
Vibration from blasting	●		

Notes

× Usually secondary parameter

○ Frequently important parameter

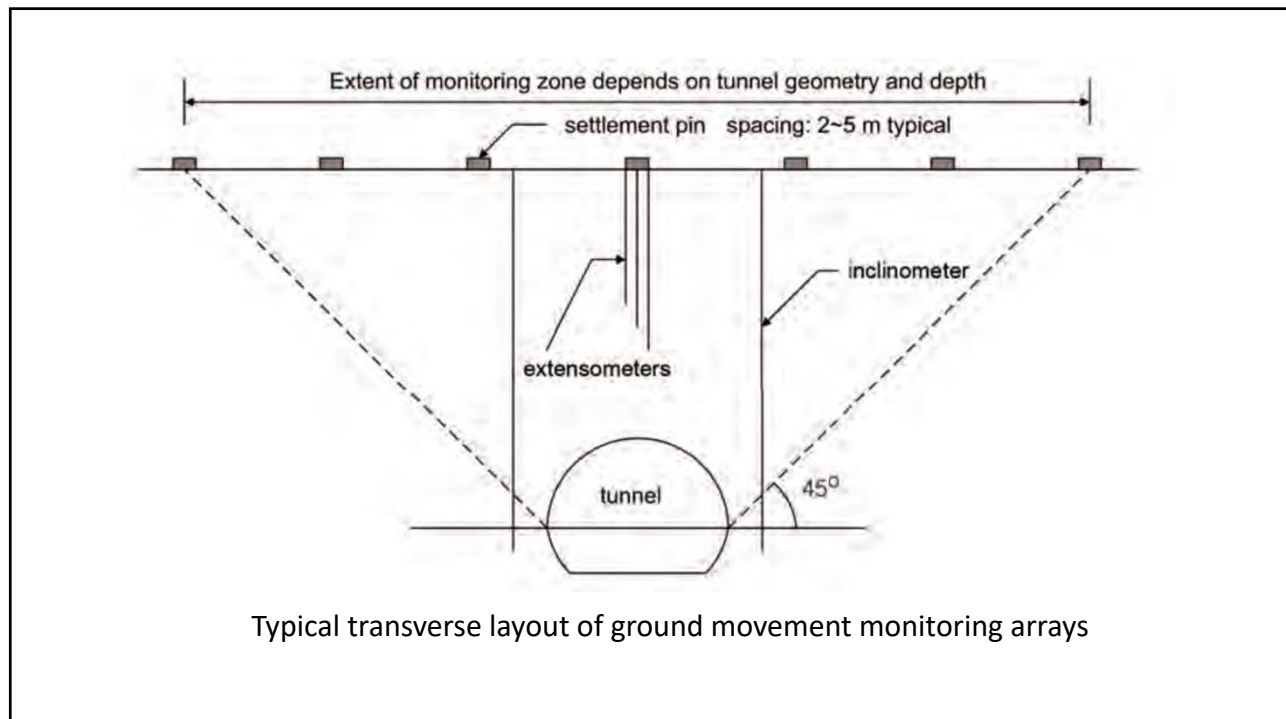
● Essential parameter, always monitored

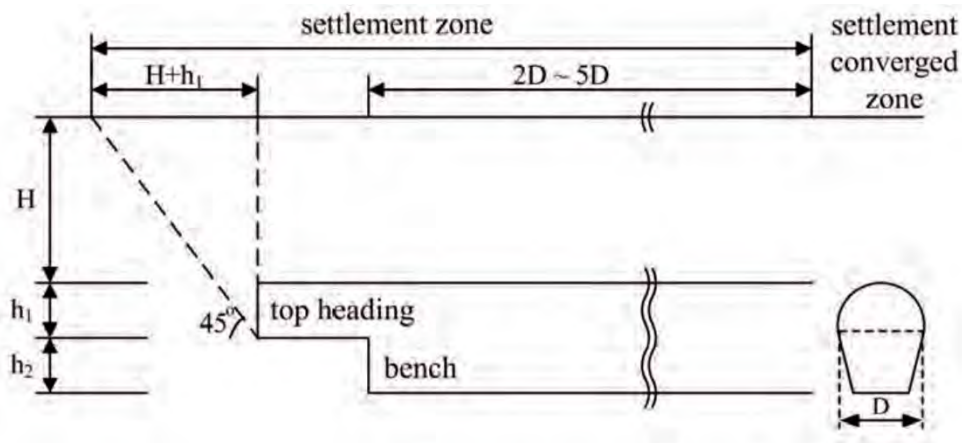
■ Essential parameter, always monitored when advance full face with pre-confinement

Table. Leading parameters to be monitored in four tunnel configurations

Objective	Instrumentation	Range	Resolution	Accuracy
Extrusion of ground ahead of face	Increx probe	0.1 mm	0.01 mm	±0.003 mm/m
	Sliding micrometer	1 m	0.01 mm	0.002 mm/m
	Sliding deformeter	1 m	0.01 mm	0.02 mm/m
Relative vertical movement	Precise leveling pins installed on structures, settlement points, geodetic surveying targets in structures or tunnel linings	any	0.1 mm	0.5–1.0 mm
	Precise liquid level settlement gauges with LVDTs installed in surface structures	100 mm	0.01–0.02 mm	±0.25 mm
	Borehole magnet extensometer	any	±0.1 mm	±1–5 mm
	Borehole rod or invar tape extensometers	100 mm	0.01 mm	±0.01–0.05 mm
	Satellite geodesy	any	to ±50 mm	to ±1 mm
Lateral displacement	Surface horizontal invar wire extensometers	0.01%	0.001–0.005%	0.01–0.05 mm
Change in inclination	Borehole electrolevels; Electrolevel beams on structures and in tunnels; 'tilt meters'	50 mm/m (to 175 mm/m)	0.05 mm/m (to 0.3 mm/m)	to 0.1 mm/m
	Horizontal borehole deflectometer	±50 mm	±0.02 mm	±0.1 mm
	Borehole inclinometer probes	±53° from vertical	0.04 mm/m	±5 mm/25 m
Change in earth pressure	'Push-in' total pressure cells	up to 1 MPa	up to 0.1% FS	up to 1.0% FS
Change in water pressure	Standpipe piezometers	any	±10 mm	±10–20 mm
	Pneumatic piezometer (pore pressures are balanced by applied pneumatic pressures); Electronic (vibrating-wire type) piezometric sensors	0–20 bar	0.01 bar	0.5% FS ± 0.02 bar
Crack or joint movement	Telltales	±20 mm	0.5 mm	±1 mm
	Calliper pins/micrometer, or mechanical strain gauges	up to 150 mm	0.02 mm	±0.02 mm
	Vibrating wire jointmeters	up to 100 mm	up to 0.02% FS	up to 0.15% FS
Strain in structural member or lining	Vibrating wire strain gauges	up to 3000 µε	0.5–1.0 µε	±1–4 µε
	Fibre optics	to 10,000 µε (1% strain)	5 µε	20 µε
Tunnel lining diametrical distortion	Tape extensometers across fixed chords	up to 30 mm	0.001–0.05 mm	±0.003–0.5 mm
	3D geodetic optical leveling ('retro' or 'botfox') targets, leveling diodes or prisms	any	0.1–1.0 mm	0.5–2.0 mm
	Strain gauged borehole extensometers installed from within tunnel	100 mm (3000 µε)	0.01 mm (0.5 µε)	±0.01–0.05 mm (±1–10 µε)
	Convergence system	±50 mm	0.01 mm	±0.05 mm
	Total pressure (or 'stress') cells	2–20 MPa	0.025–0.25% FS	0.1%–2.0% FS
Lining stress	Flow meter	any	1 litre/min	2 litre/min
Lining leakage	Triaxial vibration monitor/seismograph	250 mm/sec	0.01–0.1 mm/sec	3% at 15 Hz
Vibration				

Types of instrumentation equipment (modified Tunnel Lining Design Guide, 2004)





Typical longitudinal layout of ground movement monitoring arrays

GROUND MOVEMENT MONITORING

Various modes of instrument operation exist, namely:

Mechanical:

- often provide the simplest, cheapest and most reliable methods of detection, transmission and readout.
- Mechanical movement detectors use a steel rod or tape, fixed to the rock at one end, and in contact with a dial gauge or electrical system at the other.
- The main disadvantage of mechanical systems is that they do not lend themselves to remote reading or to continuous recording.

Optical:

- used in conventional, precise and photogrammetric surveying methods of establishing excavation profiles, measuring movements of excavation boundaries and monitoring surface subsidence.

GROUND MOVEMENT MONITORING

Hydraulic and Pneumatic:

- diaphragm transducers that are used for measuring water pressures, support loads and so forth.
- The quantity measured is a fluid pressure which acts on one side of a flexible diaphragm made of a metal, rubber or plastic.

Electrical:

- the most common instrument mode used in mines, although mechanical systems still find widespread use in displacement monitoring.
- Electrical systems operate on one of three principles, electric resistance strain gauge, vibrating wire and self-inductance.

Most commonly monitored variables include

- **Movement** (using surveying methods, surface devices such as crack gauges and tape extensometers, borehole devices such as rod extensometers or inclinometers);
- **Rock stresses** (absolute stress or stress change from borehole devices);
- **Pressure, Load and Strain** on ground support devices (e.g., load cells);
- Seismic events and Blast vibrations.

GROUND MOVEMENT MONITORING

- out of the two principle monitoring technique available, i.e, Deformation/strain and load/stress, the displacement and load have proven to be more useful, can be measured directly and monitored continuously with a relative ease.
- Stress/strain, on the other hand, must be determined indirectly from other measurements and is difficult to monitor continuously.
- Also, displacement measurements provide information on overall movement of the rock mass within the measured distance and thus do not display a large variability such is apparent when a quantity is measured at one point, such as stress or pressure (Bieniawski,1984).

GROUND MOVEMENT MONITORING

- Most electronic instrumentation measurement methods consist of three components: a transducer, a data acquisition system, and a linkage between these two components.
- A transducer is a component that converts a physical change into a corresponding electrical output signal.
- Data acquisition systems range from simple portable readout units to complex automatic systems.

Generally four types of sensors are used for all strata monitoring instruments; which are based on the following four principles;

- Linear variable differential transformer (LVDT)
- Strain gauge
- Rheostat/variable resistance
- Vibrating wire

GROUND MOVEMENT MONITORING

- Out of these four types of sensors, the first three types of sensors provide **analogue** information while the last one sends only frequency which is a fixed value like digital signal.
- It is very difficult to carry the **analogue signals** of the first three types of sensors to a distant safe place without noise and distortion.
- Frequency signal is exceptionally immune from cable effects including length (several kilometers), splicing, resistance, noise pickup and moisture
- The features of vibrating wire sensors makes it superior to other sensors because it always desirable to monitor instruments from distant and safe place during underground mining.

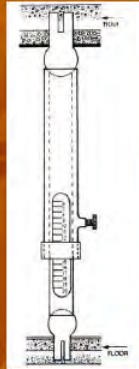
The ingredients of a successful monitoring programme should incorporate, among others, the following :

- Clear and specified objective;
- Properly designed instrumentation program;
- Installation of instruments by skilled personnel;
- Adequate data recording frequency; and
- Proper analysis of collected data.

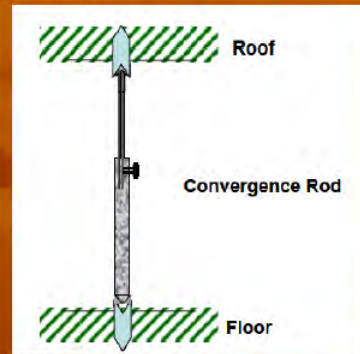
Underground

Strata Monitoring

Roof to Floor Convergence Measurement



Telescopic Convergence Rod



Principle : It is a simple instrument consisting of a graduated rod fitted in a pipe. It has a least count of 0.5 to 1 mm, and the telescopic movement is for a length of 2 to 4 m. The measuring points ('reference stations') are metal rods grouted in the roof and floor.

Procedure : Measurements are taken by simply stretching the telescopic rod between the reference points, and reading the graduations on the rod. These indicators are useful for understanding the roof to floor closure in the advance galleries at various stages of extraction.

Underground

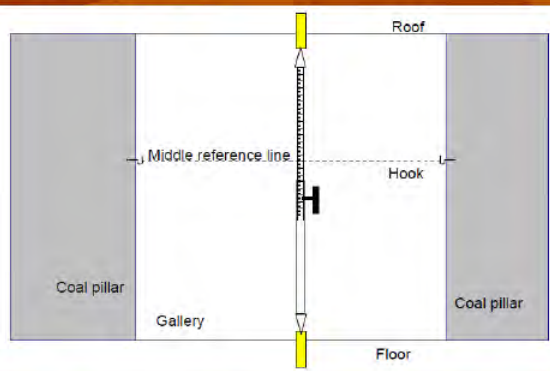
Strata Monitoring

Roof to Floor Convergence Measurement

Anchors placed in Roof and floor vertically in a straight line

Distance between anchors is measured with convergence rod

Sometimes hooks are placed in the both sides of the gallery in the pillars to facilitate measurement of roof convergence as well as floor heaving



Telescopic Convergence Rod

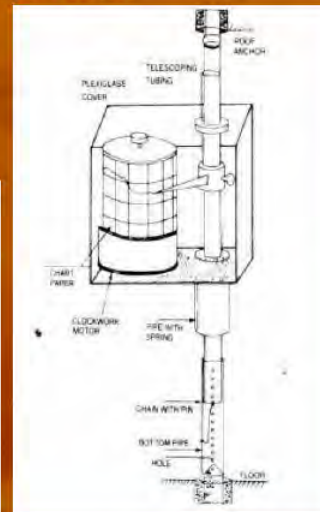
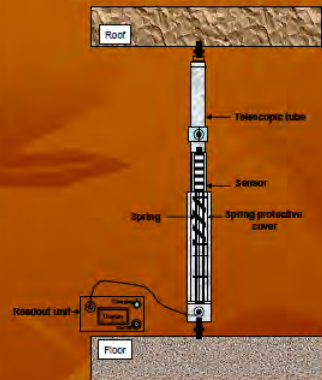


Chart Type Convergence Recorder

Underground

Strata Monitoring

Roof to Floor Convergence Measurement



- For monitoring convergence in goaf
- Sensor – Vibrating wire type or Potentiometric type
- Needs laying of wire from instrument location to measurement site
- Fails immediately with the failure of immediate roof

Remote Convergence Indicator

The RCI consists of two telescopic tubes whose ends are fixed in the roof and the floor. It has a potentiometer connected to the top sliding telescopic tube, and a slider fixed to the bottom tube. Due to the movement of the roof, the slider moves on the resistance coil, resulting in change in resistance, which is directly proportional to the amount of convergence or divergence. The change in resistance is measured with the help of a battery-operated read out unit. The RCIs were installed at the junctions in the middle of the panel.

Underground

Strata Monitoring

Convergence Measurement

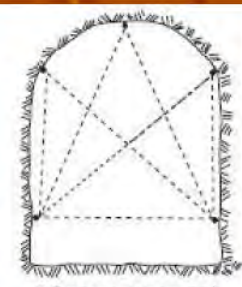


Tape Extensometer

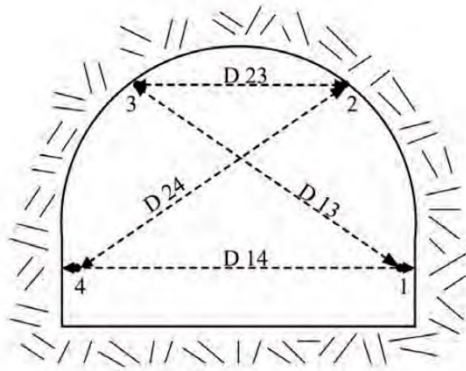
- Can be used for the measurement of convergence just like telescopic convergence rod
- Also used for surface profile movement measurement of roadways



Anchor



Reference points grouted into drill holes in rock.



Pins to measure the convergences in a tunnel
(Hudson 1995)

The convergence can be defined: $C(t) = D_o - D(t)$

Where $C(t)$ is the convergence at time t between any two opposite points in a tunnel cross section, i.e., points 1 and 3, 2 and 4, etc. in the above Figure. D_o is the initial measurement of the distance between the two points, and $D(t)$ is the measurement at time t of the distance between the two points. The convergence is considered positive if the distance between the two points has decreased and negative if it has increased.



Measuring results are absolute
Accuracy 1-2 mm
Information about deformation of the tunnel walls surface
Information about the cavity deformation of the tunnel walls surface
Lower accuracy is compensated by absolute measuring results

Illustration of optical 3D convergence measurement

Underground

Strata Monitoring



Tell tale Extensometer

Tell-tales are a low cost, easily installed monitoring device which will provide a Continuous Visual Indication of the roof Conditions

For monitoring

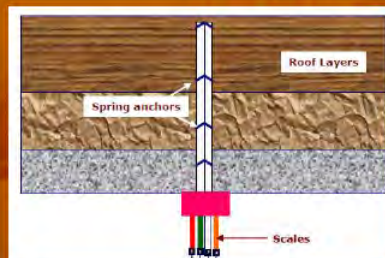
- Immediate Roof convergence
- Bed separation between layers
- Progressive failure height of the strata
- Ensure efficacy of bolting and influence of extraction or development
- Fixed only in the roof of the gallery
- Mechanical type
- Manual reading of deformation
- Maximum height can be fixed is 10 m or longer

Underground

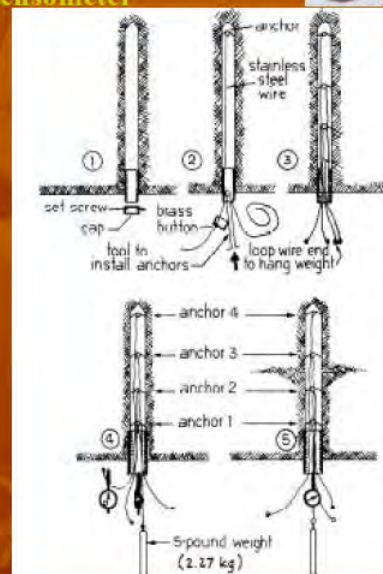
Strata Monitoring



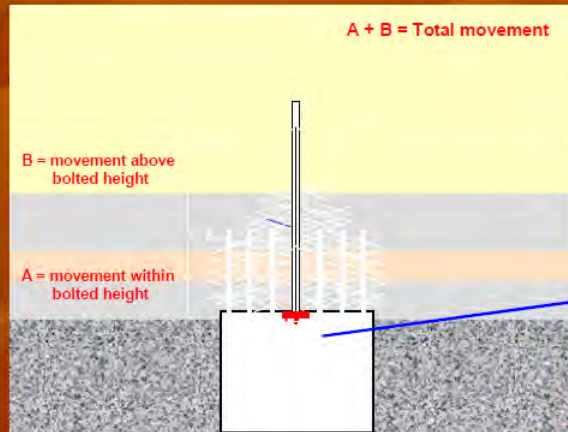
Spring type Tell tale Extensometer



Multi wire Telltale



Safety Monitoring



Colour	Graduation
A Red	60mm
B Red	55mm
TOTAL	115mm

A

B

Underground

Strata Monitoring



ACTION LEVELS

GREEN 0 - 25 mm

YELLOW 25 - 50 mm

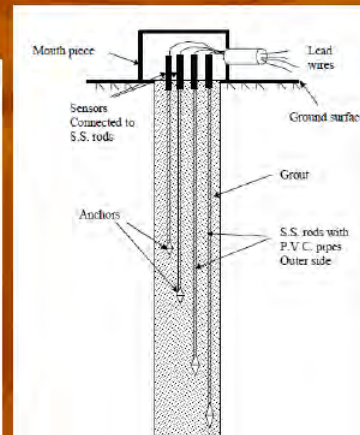
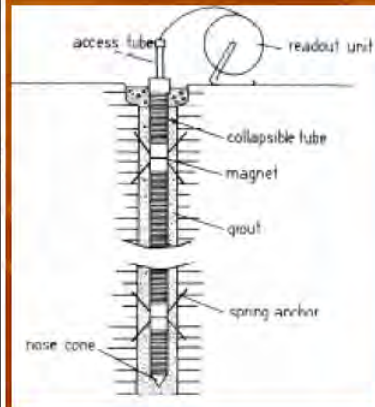
RED 50 mm +

Restrict access.
Investigate.
Install longer reinforcement

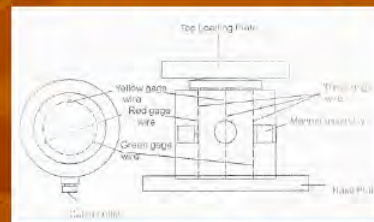
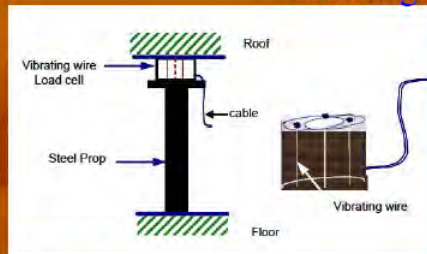
- Mechanical fixing reading with the use of VW or resistance or magnetic anchors



Multipoint Extensometers



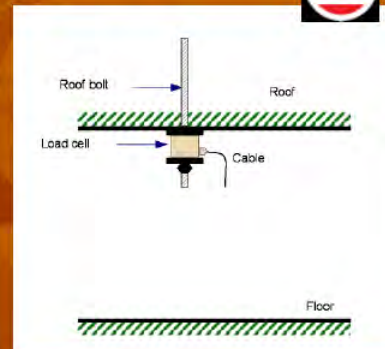
Vibrating Wire Load cells



- The load cell is a transducer working on **vibrating-wire principle**.
- It has three stretched wires housed in a metal cylinder, which are plucked by an electric pulse of high energy.
- Changes in the load exerted on the cell cause changes in the length of the wire, resulting in variations of frequency of vibration of the wire.
- As the load increases the frequency decreases accordingly and vice-versa.
- This frequency is measured by a digital read-out unit, and is converted into load using calibration charts. Efficacy and adequacy of the support system can be inferred on the basis of these load cells.



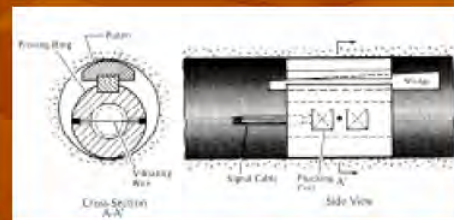
ROCKBOLT LOAD CELL



- The load cell is a transducer working on **vibrating-wire principle**.
- The load cells should be installed under the roof bolts using specially prepared steel seating arrangement .
- The load cells should be installed at the junctions to understand the change in load over the roof bolt during the operation.



Stress Cell

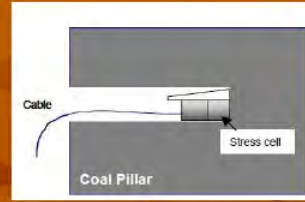


Principle : These are used for measuring **unidirectional stress change** in the pillar. It consists essentially of a wire tensioned across a steel cylinder (38 mm diameter). As the stress within the rock/coal changes, the cylinder deforms, causing tension in the wire to change.

Procedure : A bore hole of 38 mm diameter is required for installing the stress meters, preferably at mid height of the pillar either horizontally or slightly rising/dipping according to dip of the seam. The stress meter along with wedge and platen assembly is set in the borehole with the help of special installation tools, at a depth of about 4 to 9 m.

Underground

Strata Monitoring



Installation Coal Pillar

Underground

Strata Monitoring



STRESS CELL – 3 DIMENSIONAL MEASUREMENT

Stress cell consists of a soft inflatable membrane with 18 electrical **resistance** strain gauges mounted flush on its outer surface. These gauges are glued to the surface of the borehole to directly measure any strain changes that occur in the rock.



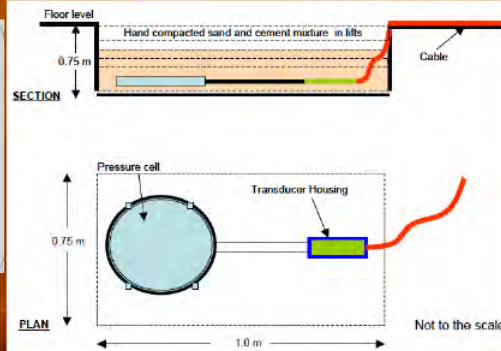
Underground

Strata Monitoring



Earth Pressure Cell

- The pressure cell is also based on the principle of **vibrating-wire** technology.
- It is constructed from two stainless steel plates welded together around the periphery so as to leave a narrow space between them.
- This space is completely filled with de-aired hydraulic oil (Glycol).
- It is connected to a vibrating-wire pressure transducer where the oil pressure is converted to an electrical signal which is transmitted through a signal cable to the readout unit.
- The vibrating-wire based pressure cell has the advantage of long-term stability and remote monitoring with cables.



Underground

Monitoring



Borehole Pressure Cells

Borehole Pressure Cells have a long term track record of stress monitoring in both elastic and viscoelastic rock. While typically classified as a soft inclusion, the cell may be filled with mercury to gain stiffness for use as a rigid inclusion in low modulus rock. Product development is largely the result of research done by the U.S. Bureau of Mines (Panek & Stock 1964, and Smith 1972).

Cells are available in two basic configurations, a miniature flatjack version (BPC) and a cylindrical pressure cell (CPC). Due to the flat design, the BPC responds primarily to the stress in the plane perpendicular to the cell, and is only slightly affected by stress in the same plane. Two BPC's mounted at right angles to each other in the same borehole therefore will monitor the principal stresses in the plane perpendicular to the borehole. Biaxial stress measurement will require three BPC's in the same borehole.

While not appropriate for anisotropic stress conditions, the CPC will measure the average change in the principal stresses in the plane perpendicular to the borehole.



Both the BPC and CPC are two, either stainless or copper plates, welded together around their periphery. The space between the plates is filled with deaired fluid, and leads to the borehole collar via a high-pressure stainless tube. Attached to this tube is a check valve and hydraulic coupler to facilitate pumping up the cell, and either a gauge or pressure transducer type readout.

In a typical installation, the BPC is grouted into the borehole. Once the grout has set, the cell is pressurized to slightly over the estimated stress. The hydraulic pump is then disconnected, with the pressure being held by the integral check valve. The system is then allowed to stabilize. In elastic rocks, given the modulus of the rock, the stress change may be determined by the corresponding change in hydraulic pressure.

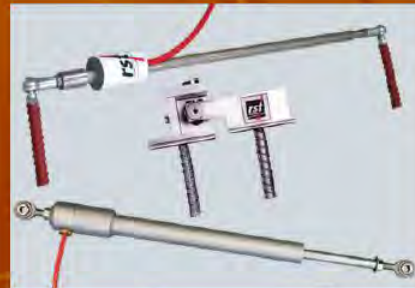
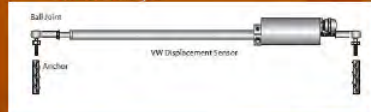
In rocks that exhibit plastic creep, the cell will not only measure stress change, but the final equilibrium pressure approximates the insitu stress.

Underground

Strata Monitoring

Crack meter

- Monitoring joints for unexpected movement to provide early warning of performance problems.
- Monitoring joints and cracks in structures that may be affected by nearby excavation and construction activities.
- Monitoring cracks in structures that have experienced seismic activity.



Underground

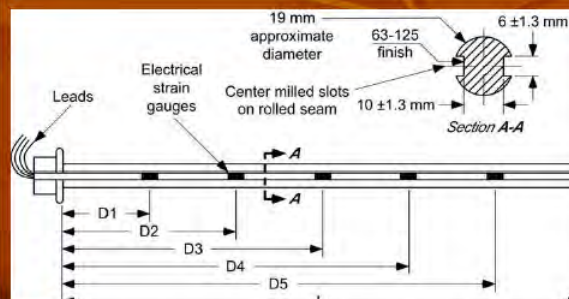
Strata Monitoring

ROCKBOLT STRAIN GAUGE

- Strain gauged rock bolts can be used to measure the distribution of bolt loads for design purposes.
- Bolt loads can be monitored along each bolt by the use of multiple strain gauges.
- Multiple pairs of strain gauges are embedded at regular spacings along each bolt enabling the axial loads and bending moments to be measured along its full length.
- strain gauged rockbolts provide a valuable tool for monitoring support system performance against design specification.



- Axial loads and strains
- Bending loads and strain difference
- High strain zones from contour plots

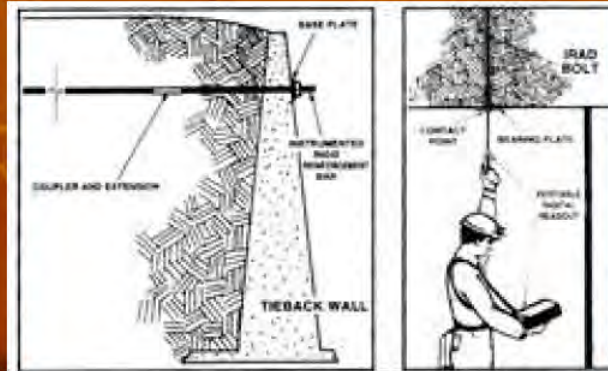


ROCKBOLT STRAIN GAUGE

➤ Rockbolt strain gauge can be fixed in roof, sides, surface slopes, embankments, etc.

➤ Wireless local reading or remote hard-wire reading up to two kilometres away are available.

➤ Normally, for the 22 mm diameter bolts, the bolt loads should not exceed 25 t (250 kN), which is the yield strength of the bolt steel.



S.No	INSTRUMENT	PARAMETER MEASURED	ACCURACY
1.	Telescopic Convergence Indicator	Roof to floor Convergence	0.5 to 1 mm
2.	Remote Convergence Indicator	Roof to Floor Convergence	0.5mm
3.	Load cell	load over the support or Rockbolt load	0.001 Tonne
4.	Stress Cells	Change in stress over pillar or 3D stress	0.5% of full Scale
5.	Earth Pressure Cell or Borehole Pressure Cell	Stress over pillar	1%
6.	Rockbolt Strain gauge	Load profile of Rockbolt	
7.	Tell tale extensometer	Convergence & Bed Separation in the immediate roof	1mm
8.	Multi Point Borehole Extensometer (Sonic or Magnetic)	Bed separation, Caving behaviour (Goaf)	± 0.1mm
9.	Pillar Strain meter or Crack meter	Deformation of pillar or crack growth	0.5mm
10.	Continuous Pressure Chart Recorder	Monitoring leg pressure of chock shield	1.5%

Types of Instrumentation for Ground Monitoring

Measurement	Suitable Monitoring Instruments
Surface deformation	Surveying methods, Crack gages, Tilt meters, Multi-point liquid level gauges
Sub surface deformation	Inclinometers, Fixed borehole extensometers, slope extensometers, shear plane indicators, multiple deflect meters, in-place inclinometers, combined inclinometer-piezometer system, Acoustic emission monitoring
Ground water pressure	Single or multi-point piezometers, combined piezometer and inclinometer systems

SLOPE MONITORING INSTRUMENTS

Extensometers

Time domain reflectometry (TDR)

Inclinometers

Piezometers

Crack Meters

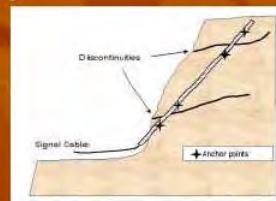


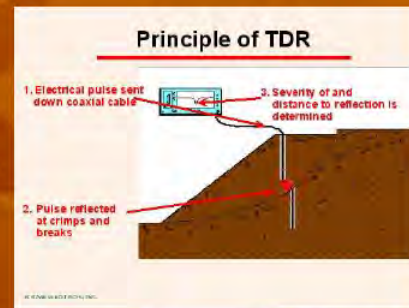
Fig: slope with Extensometer

Extensometers

Borehole extensometers consists of tensioned rods anchored at different points in a borehole. Changes in the distance between the anchor and the rod head provides the displacement information for the rock.

Time Domain Reflectometry

- * lower installation costs
- * no limits on hole depth
- * immediate determination of movement
- * remote data acquisition capability

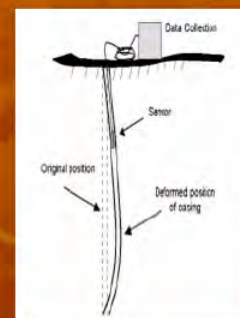


In TDR, a cable tester sends a voltage pulse waveform down a cable grouted in a borehole. If the pulse encounters a change in the characteristic impedance of the cable, it is reflected. This can be caused by a crimp, a kink, the presence of water, or a break in the cable. The cable tester compares the returned pulse with the emitted pulse, and determines the reflection coefficient of the cable at that point. The change in impedance with time corresponds qualitatively to the rate of ground movement.

Inclinometers

inclinometers is important to slope stability studies for the following reasons (Kliche, 1999)

- to locate shear zone(s);
- to determine whether the shear along the zone(s) is planar or rotational;
- to measure the movement along the shear zone(s) and determine whether the movement is constant, accelerating, or decelerating.



Monitoring slopes and landslides to detect zones of movement
Monitoring dams, dam abutments, and upstream slopes.
Monitoring the effects of tunneling operations

Piezometers

- Monitoring pore water pressures to determine safe rates of fill or excavation.
- Monitoring pore water pressures to evaluate slope stability.
- Monitoring dewatering systems used for excavations.
- Monitoring ground improvement systems, such as vertical drains and sand drains.
- Monitoring pore pressures to check the performance of earthfill dams and embankments.
- Monitoring pore pressures to check containment systems at landfills and tailings dams.

- Vibrating wire
- Pneumatic
- Standpipe piezometers



Crack Meters



Crack meters can be very useful tools in the early detection of deforming mass movements. These devices measure the displacement between two points on the surface that are exhibiting signs of separation.